Determination of Drought Triggering Probability via Bayesian Network

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Abstract

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1 Introduction

Drought Anticipatory Action (DAA)

bnet (Bayesian network) (see Ref.[3])

Standardized Precipitation Index (SPI) ranges from -4 (least rain) to 4 (most rain).

2 Notation and Conventions

We will use $\mathbb{1}(S)$ to denote the indicator function (a.k.a. the truth function). $\mathbb{1}(S)$ equals 1 if statement S is true, and 0 if S is false. For example, $\mathbb{1}(x \geq 0)$ equals 0 if $x < 0$ and 1 otherwise.

In this article, random variables will be indicated by underlining. For example, $P(\underline{x} > x)$ will denote the probability that the random variable $\underline{x}$ is greater than x.

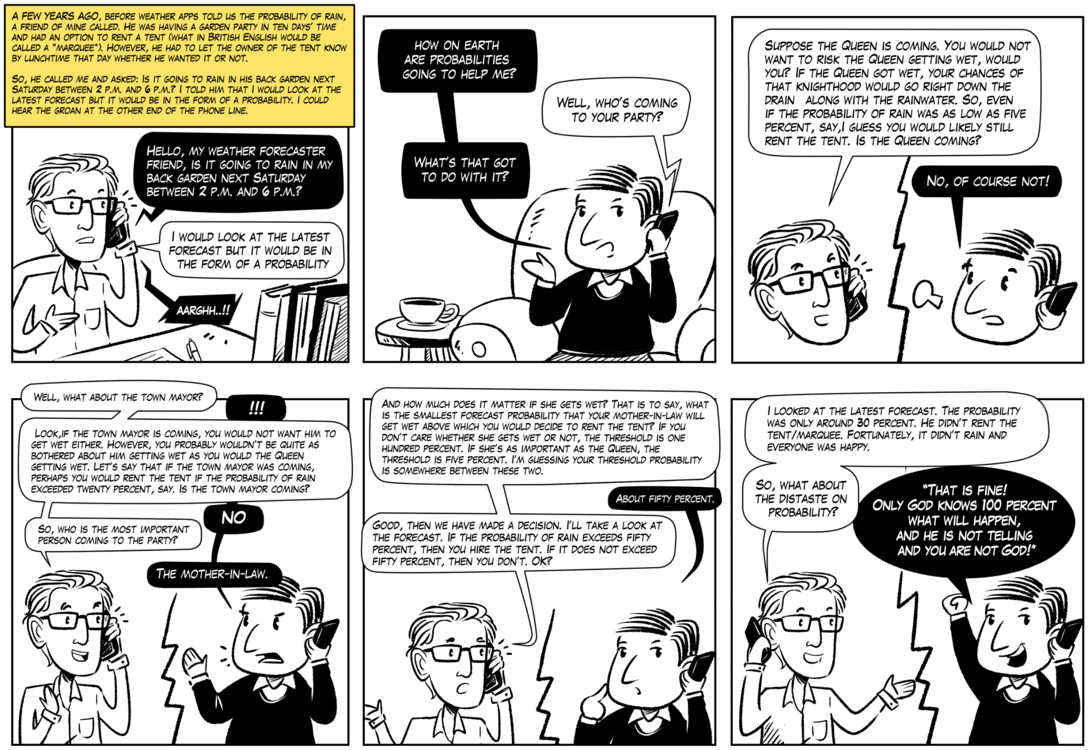
Let

$N =$ the number of times (seasons) in which predicted and observed droughts were compared.

$N_{x|a} =$ number of events in which a number of $a \in \{0,1\}$ droughts occurred and a number of $x \in \{0,1\}$ droughts were predicted. For example, $N_{1|0}$ is the number of events in which no drought actually occurred but it was predicted nonetheless (False Alarm).

$N_{0|0} =$ **True Negatives** (TN), correct rejection

$N_{1|1} =$ **True Positives** (TP), hit



Source: Palmer, Tim. The Primacy of Doubt: From climate change to quantum physics, how the science of uncertainty can help predict and understand our chaotic world. Oxford University Press, 2022.

Figure 1: Illustration from Tim Palmer's book Ref.[2]

$N_{0|1}$ = **False Negatives** (FN), miss, type II error or underestimation
$N_{1|0}$ = **False Positives** (FP), false alarm, type I error or overestimation

$$N_{|a} = \sum_{x'} N_{x'|a}, \quad N_{x|} = \sum_{a'} N_{x|a'} \tag{1}$$

$$N = \sum_{x} \sum_{a} N_{x|a} = \sum_{a} N_{|a} = \sum_{x} N_{x|a} \tag{2}$$

$$P(x|a) = R_{x|a} = \frac{N_{x|a}}{N_{|a}}, \ P'(a|x) = R'_{a|x} = \frac{N_{x|a}}{N_{x|}} \tag{3}$$

$$\sum_{x} P(x|a) = \sum_{a} P'(a|x) = 1 \tag{4}$$

Note that both $P(x|a)$ and $P'(a|x)$ are conditional probability distributions.

Note that $N_{x|a}$ for $x, a \in \{0, 1\}$ yields 4 degrees of freedom (dofs). Instead of those 4 dofs, we will use N and 3 others dofs, namely, hit rate (HR), false alarm rate (FAR) and area under ROC curve (AUROC). HR, FAR and AUROC are defined in Appendix A

3 DAA bnet

Let

$\kappa = (\sigma, \delta)$ = the drought kind, where $\sigma \in \{\text{MAM, JJAS}\}$[1] is the season, and $\delta \in \{\text{Moderate, Extreme, Severe}\}$ is the drought degree.

S_κ = **Severity** of drought is defined as minus SPI (higher rain, higher SPI, less severe drought). For flood (FAA), we would use $+SPI$ instead.

$P(S_\kappa)$ = likelihood, probability that a drought of severity S_κ will occur in the near future (weeks to months), based on forecasts.

μ = ensemble member index

M = number of ensemble members, 51 for $ECMWF$ (European Centre for Medium-Range Weather Forecasts)

$\sigma \backslash \delta$	Moderate	Extreme	Severe
MAM	−0.55	−0.98	−0.99
JJAS	−0.41	−0.98	−0.99

Table 1: SPI threshold for Karamoja, Uganda. Cells give $-S^*_\kappa$, where $\kappa = (\sigma, \delta)$

S^*_κ = severity threshold for drought (See Table 1 for examples of S^*_κ.)

$$P(\underline{S}_\kappa \geq S^*_\kappa) = \frac{1}{M} \sum_{\mu=1}^{M} \mathbb{1}(S^{\kappa,\mu} \geq S^*_\kappa) \tag{5}$$

Let $X^i_\kappa \in \{0, 1\}$ for $i = 1, 2, 3, \ldots, nx$ represent a flag for local problems such as high road density, high population density, historical catastrophic events, etc. It equals 1 if a particular danger is present, and 0 if not.

Let

$\epsilon \in \mathbb{R}$ be a normally distributed random variable with mean zero and some a priori given deviation.

RC^i_κ = replacement cost for flag X^i_κ

IC^i_κ = insurance (i.e., preventive) cost for flag X^i_κ

Note that insurance always costs more than replacement, so

$$0 \leq IC^i_\kappa \leq RC^i_\kappa \tag{6}$$

$P^i_{danger,\kappa} = \frac{IC^i_\kappa}{RC^i_\kappa}$ probability of danger for flag X^i_κ.

Note that less danger means less insurance cost.

P^*_κ = probability threshold for drought

$$P^*_\kappa = f_\kappa(X^1_\kappa, X^2_\kappa, \ldots, X^{nx}_\kappa) + \epsilon \tag{7}$$

where we model the function $f_\kappa()$ by

$$f_\kappa(X^1_\kappa, X^2_\kappa, \ldots, X^{nx}_\kappa) = 1 - \max_i \{P^i_{danger,\kappa} X^i_\kappa\} \tag{8}$$

[1] Letters here stand for the first letter of the months of the year.

TD_κ = triggering decision to do drought warning
P^*_κ = threshold probability that triggers a drought warning

$$TD_\kappa = \mathbb{1}\left[P(\underline{S}_\kappa \geq S^*_\kappa) > P^*_\kappa\right] \tag{9}$$

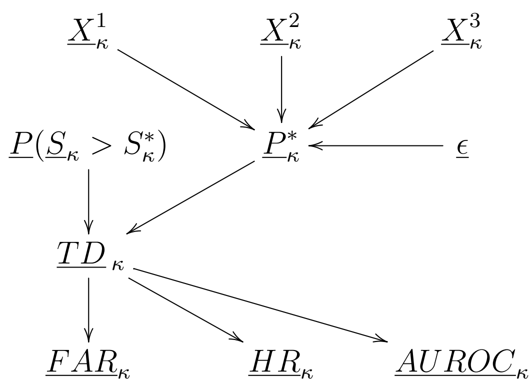


Figure 2: DAA bnet. We use one of these bnets for each κ value and for each grid cell of 0.25 degrees in latitude by 0.25 degrees in longitude. 1 degree in latitude or longitude is approximately 100 km.

The TPMs[2] printed in blue, for the bnet Fig.2, are as follows.[3]

$$P^*_\kappa = f_\kappa(X^1_\kappa, X^2_\kappa, \ldots, X^{nx}_\kappa) + \epsilon \tag{10}$$

$$TD_\kappa = \mathbb{1}[P(\underline{S}_\kappa > S^*_\kappa) > P^*_\kappa] \tag{11}$$

$$P(Y^i_\kappa | TD_\kappa) = \text{to be learned empirically from dataset} \tag{12}$$

where $(Y^1_\kappa, Y^2_\kappa, Y^3_\kappa) = (FAR_\kappa, HR_\kappa, AUROC_\kappa)$.

Algorithm for finding best triggers T^*_κ:
Enter bnet Fig.2 into bnet software PyAgrum (Ref.[1]). Enter into software available evidence for nodes $P(\underline{S}_\kappa > S^*_\kappa)$, (X^1, X^2, X^3) and $(Y^1_\kappa, Y^2_\kappa, Y^3_\kappa)$. Allow software to infer the distribution of P^*_κ. Choose for P^*_κ the state with maximum probability.

[2]TPM=Transition Probability Matrix. This is the same thing as CPT=Conditional Probability Table

[3]A deterministic node satisfying the structural equation $y = f(x_1, x_2)$, has the TPM: $P(y|x_1, x_2) = \delta(y, f(x_1, x_2))$, where $\delta(a, b) = \mathbb{1}(a = b)$ is the Kronecker delta function.

4 Relation to Climada

This project used the Climada (see Ref.[6]) software library. Table 2 relates their terminology to ours.

In Climada, a hazard model yields the intensity for a particular event.

Climata terminology	Our terminology
Intensity of event	S=severity of event
Vulnerability= $\frac{Impact}{Exposure}$	$P_{danger} = \frac{IC}{RC}$=danger probability
Impact	IC= insurance cost ($)
Exposure	RC=replacement cost ($)

Table 2: Relationship between Climata terminology and ours

Climada takes as input the intensity of an event, and calculates from this the event's vulnerability, impact and exposure. In this project, we use Climada for each flag X^i.

5 Risk

$$\begin{aligned} Risk_\kappa &= E[\underline{IC}_\kappa] = \frac{1}{M}\sum_{\mu=1}^{M} IC^{\kappa,\mu} && (13)\\ &= \sum_{IC_\kappa} P(IC_\kappa) IC_\kappa && (14)\end{aligned}$$

A **risk matrix** is a 4×4 matrix with X axis equal to insurance cost (IC) (a.k.a. impact) and Y axis equal to likelihood P, with IC binned into four values and P binned into 4 values. The matrix elements give the value of some arbitrary function $f(IC, P)$ defined as risk. Usually, $f(IC, P) = IC * P$ but not necessarily. I find risk matrices arbitrary and not very useful. The Wikipedia article on risk matrices (Ref.[5]) agrees.

A Appendix: Functions of $N_{x|a}$

Terminology Table Adapted from Wikipedia Ref.[4]

positive events (P): $P = N_{|1} = \sum_x N_{x|1}$, the number of real positive events in the data

negative events (N): $N = N_{|0} = \sum_x N_{x|0}$, the number of real negative events in the data

sensitivity, recall, true positive rate (TPR), or hit rate (HR):

$$TPR = R_{1|1} = \frac{N_{1|1}}{N_{|1}} = \frac{N_{1|1}}{N_{1|1} + N_{0|1}} = 1 - R_{0|1} \quad (15)$$

specificity, selectivity or true negative rate (TNR):

$$TNR = R_{0|0} = \frac{N_{0|0}}{N_{|0}} = \frac{N_{0|0}}{N_{0|0} + N_{1|0}} = 1 - R_{1|0} \tag{16}$$

precision or positive predictive value (PPV):

$$PPV = \frac{N_{1|1}}{N_{1|1} + N_{1|0}} = 1 - FDR \tag{17}$$

negative predictive value (NPV):

$$NPV = \frac{N_{0|0}}{N_{0|0} + N_{0|1}} = 1 - FOR \tag{18}$$

miss rate or false negative rate (FNR):

$$FNR = R_{0|1} = \frac{N_{0|1}}{N_{|1}} = \frac{N_{0|1}}{N_{0|1} + N_{1|1}} = 1 - R_{1|1} \tag{19}$$

fall-out, false positive rate (FPR), or false alarm rate (FAR):

$$FPR = R_{1|0} = \frac{N_{1|0}}{N_{|0}} = \frac{N_{1|0}}{N_{1|0} + N_{0|0}} = 1 - R_{0|0} \tag{20}$$

false discovery rate (FDR):

$$FDR = \frac{N_{1|0}}{N_{1|0} + N_{1|1}} = 1 - PPV \tag{21}$$

false omission rate (FOR):

$$FOR = \frac{N_{0|1}}{N_{0|1} + N_{0|0}} = 1 - NPV \tag{22}$$

accuracy (ACC):

$$ACC = \frac{N_{1|1} + N_{0|0}}{N_{|1} + N_{|0}} = \frac{N_{1|1} + N_{0|0}}{N_{1|1} + N_{0|0} + N_{1|0} + N_{0|1}} \tag{23}$$

balanced accuracy (BA):

$$BA = \frac{R_{1|1} + R_{0|0}}{2} \tag{24}$$

F1 score is the harmonic mean of precision and sensitivity:

$$F_1 = 2 \times \frac{PPV \times R_{1|1}}{PPV + R_{1|1}} = \frac{2N_{1|1}}{2N_{1|1} + N_{1|0} + N_{0|1}} \tag{25}$$

The area under the ROC (Receiver Operating Characteristic) curve, known as AUC or AUROC, is defined by Fig.3. One set of 4 values $N_{x|a}$ where $a, x \in \{0, 1\}$, only gives one point of the ROC curve. To get a full ROC curve, we need to vary a parameter. In the case of DAA, we vary geographical location, for example.

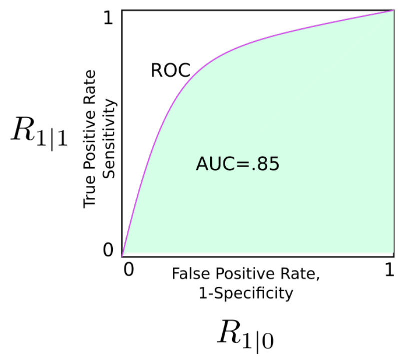


Figure 3: Example of ROC. Green shaded area is the AUC of the ROC.

References

[1] Agrumery. PyArum, free open source app at github. `https://github.com/agrumery/aGrUM`.

[2] Tim N. Palmer. *The Primacy of Doubt: From climate change to quantum physics, how the science of uncertainty can help predict and understand our chaotic world.* Oxford University Press, 2022.

[3] Robert R. Tucci. Bayesuvius (free book). `https://github.com/rrtucci/Bayesuvius`.

[4] Wikipedia. Receiver operating characteristic. `https://en.wikipedia.org/wiki/Receiver_operating_characteristic`.

[5] Wikipedia. Risk matrix. `https://en.wikipedia.org/wiki/Risk_matrix`.

[6] ETH Zurich. Climada. `https://climada-python.readthedocs.io/en/stable/`.